

US HOS and ELD Log Compliance: Team Reference Guide

Introduction

This document synthesizes findings from the provided notes, detailing the rules, monitoring processes, adjustment protocols, and key terminology related to HOS compliance and ELD logs. It serves as a comprehensive reference for all team members, including the Product Owner (PO), AI Specialist (AI), and Full Stack Developers (FS1, FS2, FS3), to guide the design and implementation of the violation detection and correction system, including the Chrome extension, admin dashboard, and AI-driven correction tools.

This guide covers:

1. **HOS Rules:** Core regulations governing driver hours, including cycle limits, shift rules, breaks, and split breaks.
 2. **ELD Log Monitoring:** Processes for tracking compliance via ELD data, including status updates, load information, and violation triggers.
 3. **Adjustment Rules:** Protocols for correcting violations in ELD logs while maintaining compliance.
 4. **Terminology:** Definitions of key terms for team-wide clarity.
 5. **Key Insights and Implications:** Summary of findings and their impact on system development, with references to relevant team tasks.
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1. Hours of Service (HOS) Rules

HOS rules, enforced by the FMCSA, regulate commercial motor vehicle drivers' working hours to prevent fatigue-related incidents. The rules define limits on driving time, on-duty time, and mandatory rest periods. Below is a detailed explanation of the HOS rules.

1.1 Core HOS Limits

- **70-Hour Weekly Cycle:**

- Drivers are limited to **70 hours of combined driving and on-duty time** within a 7-day cycle.
 - The cycle resets only after a **34-hour consecutive rest period**, which can include:
 - **Off-duty** status.
 - **Sleeper berth** status.
 - **Personal conveyance (PC)**, which counts as off-duty time (e.g., driving for personal reasons).
 - If a 34-hour reset is not achieved, drivers operate on **recap hours**:
 - Recap hours reuse hours from 7 days prior (e.g., hours used on Day 1 are available “tomorrow”).
 - Example: If a driver used 10 hours on Day 1 (Driving + On Duty), those 10 hours become available “tomorrow” without a reset. This rule keeps recurring for every following day.
 - **Related Task:** The AI Specialist is tasked with implementing weekly hour calculations for 60/70-hour rules and 34-hour restart.
- **14-Hour Shift Limit:**
 - A driver’s shift (continuously ticking down 14 hour window) cannot exceed 14 hours from the shift’s start.
 - The 14-hour clock runs continuously, and can only be extended through split-breaks.
 - **Related Task:** The AI Specialist is tasked with implementing 14-hour shift window (interval) rule.
 - **11-Hour Driving Limit:**
 - Within the 14-hour shift, drivers are limited to **11 hours of driving time**.
 - Driving time is tracked separately from other on-duty activities.
 - **Related Task:** The AI Specialist is tasked with implementing 11-hour driving rule detection.
 - **30-Minute Break Requirement:**
 - Drivers must take a **30-minute break** before completing **8 hours of driving** in a shift.
 - The break can be off-duty, sleeper berth, on-duty, or a combination, and must be uninterrupted.
 - **Related Task:** The AI Specialist is tasked with implementing 30-minute break rule detection.

1.2 Split Break Rule

The split break rule allows drivers to extend their 14-hour shift without a full 10-hour rest by taking two rest periods.

- **Split Break Requirements:**
 - Consists of **two rest periods** totaling **at least 10 hours**.
 - The **smaller rest period** must be:
 - At least **2 hours**.
 - Taken in the sleeper berth, off-duty, or combination of both.
 - The **larger rest period** makes up the remainder and always must be fully uninterrupted on sleeper berth (e.g., 7 hours SB if the smaller rest is 3 hours).
 - Example: 7 hours 45 minutes + 2 hours 15 minutes = 10 hours (valid split).
 - If the smaller rest exceeds 3 hours, it extends the shift for its full duration but it only contributes towards the 10h by maximum 3h (example: if the small break is 5h it will extend the 14h shift for the full 5h but it will be only contributing 3h for a future split so the next break will still have to be at least 7h).
 - **Related Task:** The AI Specialist is tasked with creating split calculations.
- **Impact on Shift and Driving Time:**
 - **Driving Time Earned:** Additional driving time equal to **11 hours minus driving time used between split breaks**.
 - Example: 5 hours driven between breaks yields $11 - 5 = 6$ additional driving hours.
 - **Shift Time Earned:** Shift extension equal to **14 hours minus total shift time between split breaks**.
 - Example: 8 hours of shift time between breaks yields $14 - 8 = 6$ additional shift hours.

1.3 Post-10-Hour Break Requirements

- After a **10-hour rest** (or the larger portion of a split break), drivers must:
 - Perform a **Pre-Trip Inspection (PTI)**.
 - **Sign the previous day's log** (if the break ended in a new day).
- **Related Task:** Building PTI compliance and certification tracking and implementing a reminder system.

2. ELD Log Monitoring

ELD systems automatically record driver activities to ensure HOS compliance. A **Housekeeping Agent** checks logs for violations. Below are the monitoring rules and processes.

2.1 Intermediate Status Updates

- **Frequency:** Hourly updates during **Driving Status**, including:
 - **Time, Location** (GPS), **Mileage** (odometer), and status details.
- **Purpose:** Real-time tracking for compliance verification.
- **Related Task:** FS1 is tasked with building real-time monitoring APIs.

2.2 Odometer and Location Consistency

- **Rule:** Odometer and location must remain **consistent** between the end of one driving period and the start of the next across all statuses.
 - Example: Mile 1,000 at location X at the end of driving must match the next status until next driving begins.
- **Violation Trigger:** Inconsistencies suggest errors or fraud.
- **Related Task:** Implementing an odometer, engine hours, and location matching/correction tool.

2.3 Load Information

- **Post-Pickup Requirement:**
 - A **load/document number** must be recorded in the Sahara ELD platform after a **pickup (P/u)**. (Driver can also simply write commodity)
 - **GPS** confirms departure from the pickup location.
 - Missing documents trigger a violation and prompt the ELD Support team to make driver contact.
- **In-Transit Days (No Pickup):**
 - Drivers must enter a **load/document number**.
 - Missing document numbers trigger a violation and driver contact.
 - **Trigger:** System checks **Pre-Trip Inspection (PTI)** to identify in-transit days.
- **Related Task:** Checking at PTI for missing document numbers and suggesting driver contact. FS2 is tasked with building log data fetching from Sahara.

2.4 Log Signatures

- **Requirement:** Drivers must sign previous logs.
- **Trigger:** Checked during **PTI** at shift start.
- **Violation:** Missing signatures (certification events) trigger alerts.
- **Related Task:** Checking for signatures during PTI.

2.5 Pre-Trip Inspection (PTI)

- **Requirement:** A **10-15 minute On-duty not driving** for PTI at shift start, including a **trailer number check**. (Trailer Number Should Be Checked Via Dispatch Board)
- **Violation:** Missing PTI gap or incorrect trailer number triggers an AI-suggested adjustment.
- **Related Task:** Creating a PTI dashboard and implementing PTI compliance alerts.

2.6 Driver Tracking Platform (Sahara)

- **Alvys Platform:** Load info integrated with ELD.
- **Events Monitored:**
 - **Log Out/Log In During Driving:** Flagged as **unidentified driving**.
 - **AI Suggestion:** Assign miles to the driver's log.
 - **Related Task:** The AI Specialist is tasked with implementing status correction tools to handle unidentified driving.
 - **Teleportation Event:** No ELD data due to:
 - **Unplugged Device:** Requires investigation.
 - **Detection:** Driving detected without ELD updates for **15 minutes** notifies **ELD support team**.
 - ****No Service:** Self-corrects when service resumes (there will be no teleportation if network is the issue, since the data is stored offline on the device and it will be updated as soon it connects to network again).
 - **Action:** Corrections proposed after data resumes (the difference between the end of the last known drive before network disconnection and the first correct event after network resumes should be generated based on a route between those two points. (Corrections can be matched with Samsara Data but this should be optional and if there is no other data source apply the initial route generation step between the two points)
 - **Related Task:** FS1 is tasked with implementing error handling and logging for teleportation events.

2.7 Driving, Shift, and Break Monitoring

- **Driving Status:** Exceeding **11 hours** (with seconds) triggers a violation.
- **Shift Status:** Exceeding **14 hours** (with seconds) triggers a violation. (Exceeding 14hrs. + the Split Break Extension Triggers "Shift With Split Violation")
- **Break Status:**
 - Compliance with **30-minute break**, **10-hour rest**, and **split breaks** (10+ hours total).
 - Violations flagged for non-compliance.
- **End-of-Shift Report:** Triggers a comprehensive violation check (in reality the actual end of shift cannot be predicted as it purely depends on driver choice, however a check can be triggered when the driver stops and then the shift runs out, presuming he is not supposed to drive before his time is back again so this would be end of the current shift).
- **Related Task:** Building violation detection API endpoints.

2.8 Multiple Power Events

- **Rule:** Consecutive same-type events (e.g., P/u, P/u) are flagged as errors. (Additionally, if there is no power up event prior to driving event triggers a correction)

- **Action:** AI suggests corrections (e.g., remove duplicates or change them to the correctly expected event (after a power up next power event must be power down and vice versa)).
 - **Related Task:** Implementing multi-tool coordination for such corrections.
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3. Rules for Log Adjustments

Adjustments correct violations while ensuring FMCSA compliance, prioritizing minimal disruption.

3.1 Inspection Check

- **Initial Check:** Verify if an **inspection** occurred.
 - **If Logs Were Taken:** No adjustments up to the inspection point; post-inspection driving statuses can be adjusted.
 - **If No Logs Taken:** Full shift adjustments permitted, excluding duration of inspection time.
- **Related Task:** Sahara API to monitor when logs are exported.

3.2 Fuel Stop Check

- **Priority:** Avoid moving **fuel stops**.
- **If Necessary:**
 - Suggest moving fuel stop with a reminder for the driver to wait additional time.
 - The fuel card must be switched to inactive.
 - Final correction restores the fuel stop.
- **Related Task:** EFS (or similar) fuel card transaction APIs and fuel stop location matching.

3.3 Load Information Check

- **Pickup/Delivery:** Requires a **10-minute on-duty check-in**.
- **Priority:** Adjust **empty breaks** (e.g., home time, rest areas) before pickup/delivery times.
- **Related Task:** Checking load info (PU and DEL address) through API (Alvys or similar or creating a mobile driver app where they will need to upload BOLs (possible alternative to this is also an option to upload BOLs and DOT paperwork within the ELD app)) and comparing if the log events match with it.

3.4 Speed, Mileage and Shift Analysis

- **Metrics:** Total hours, shift time, miles, and **average speed**.
- **Rule:** If speed is **under 70 mph**, suggest a **shrink adjustment**.
- **Related Task:** The AI Specialist is tasked with implementing time correction tools to support shrink adjustments.

3.5 General Adjustment Principles

- Ensure HOS compliance.
 - Prioritize non-critical events for changes.
 - Maintain an audit trail.
 - **Related Task:** FS1 is tasked with adding correction audit trails.
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4. Key Terminology

- **HOS:** FMCSA regulations for driver hours.
 - **ELD:** Electronic Log Device for automatic activity recording.
 - **70-Hour Cycle:** 70-hour limit in 8 days, reset by 34 hours.
 - **Recap Hours:** Reused hours from 7 days prior. (Driving + On-Duty)
 - **14-Hour Shift:** Maximum time interval on-duty period.
 - **11-Hour Driving Limit:** Maximum driving time.
 - **30-Minute Break:** Required before 8 hours of driving.
 - **Split Break:** Two rest periods totaling at least 10 hours.
 - **Sleeper Berth:** Rest in the vehicle's sleeping compartment.
 - **Personal Conveyance (PC):** Off-duty personal driving.
 - **Pre-Trip Inspection (PTI):** Mandatory vehicle check at shift start.
 - **Load Document:** Pickup or in-transit load documentation.
 - **Unidentified Driving:** Driving without proper login/vehicle selection.
 - **Teleportation Event:** ELD data gap.
 - **Housekeeping Agent:** Log monitoring.
 - **Shrink Adjustment:** Shortening driving time.
 - **Sahara:** Real-time tracking and logbook platform.
 - **Alvys:** Load Info Platform
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5. Key Insights and Implications for Development

5.1 Insights

- **Complex HOS Rules:** Require precise algorithms for recap hours and split breaks.
- **ELD Data Integrity:** Odometer, engine hours, coordinates, sequence number, certification events for prior logs and load document consistency is critical.
- **AI Adjustments:** Context-dependent corrections need robust logic.
- **Real-Time Needs:** Live tracking and teleportation events demand scalable APIs.
- **User Needs:**
 - ELD Support Team: Intuitive Chrome extension for violations or incoming violation notifications and standalone web-app for adjusting.
 - Fleet Managers: Dashboards for monitoring.

5.2 Implications and Related Tasks

- **Violation Detection:**
 - Algorithms for 10 violation types with second-level precision.
 - **Task:** Agent to implement violation detection logic
 - **AI Correction Tools:**
 - Tools for status, time, odometer, etc., with human review.
 - **Task:** AI Specialist to build correction agent system.
 - **Chrome Extension:**
 - Real time monitoring features – violation trigger notifications and incoming potential violations.
 - **Task:** Build popup interface and violation.
 - **Admin Dashboard:**
 - Violation lists and analytics.
 - **Task:** Build violation dashboard.
 - **Backend and APIs:**
 - Real-time ELD data and adjustment workflows.
 - **Task:** Build CSV upload and parsing API (*decode ERODS export file*).
 - **Testing and Validation:**
 - Comprehensive test scenarios.
 - **Task:** PO to create test scenarios for HOS violations.
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Conclusion

This guide provides a team-wide reference for building the US HOS Violation Detection System, addressing HOS rules, ELD monitoring, and adjustment protocols. It aligns with tasks assigned to the PO, AI Specialist, FS1, FS2, and FS3, ensuring all team members can leverage these insights for development.

Next Steps:

- Use this document to inform requirements and architecture design.
- Refine violation detection algorithms.
- Develop Chrome extension and dashboard features.
- Share with the team to align on implementation priorities.